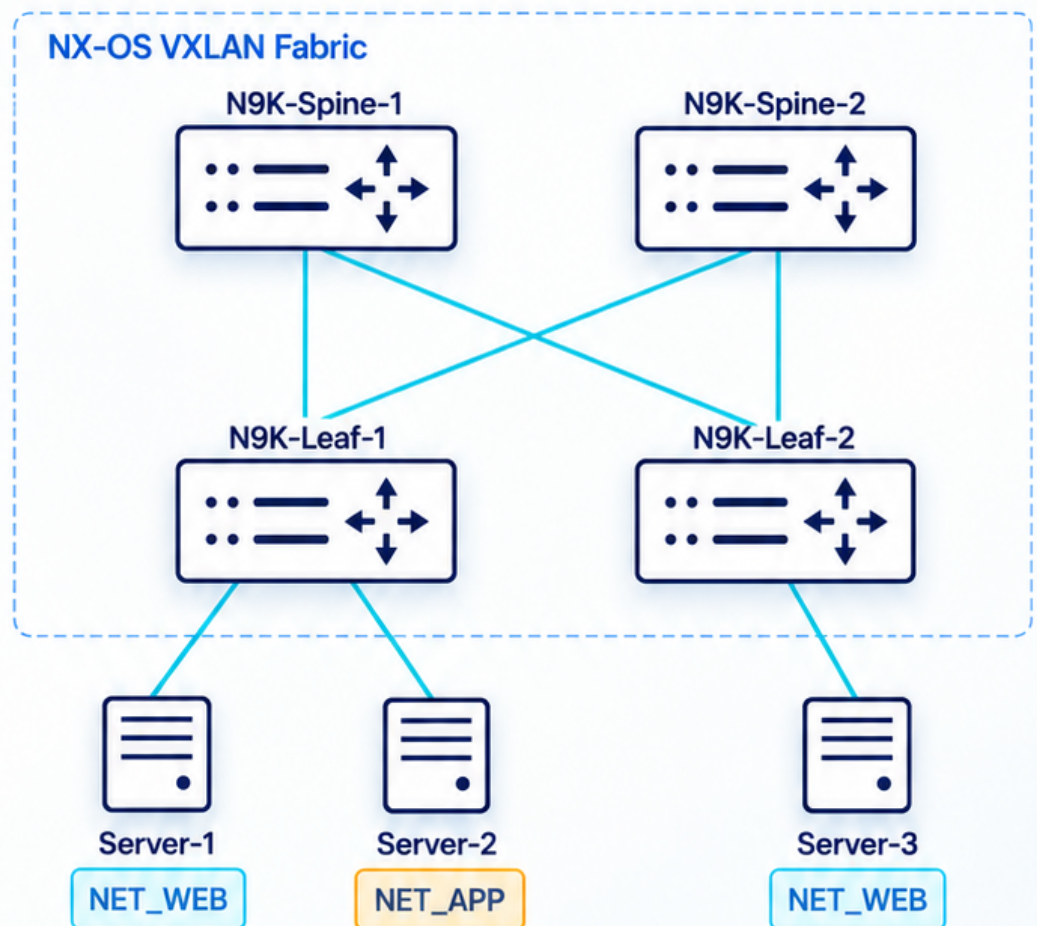


Building a VXLAN EVPN Fabric with Cisco NDFC

A Step-by-Step Introduction to Fabric Creation, VRFs, Networks and Endpoint Connectivity



Author
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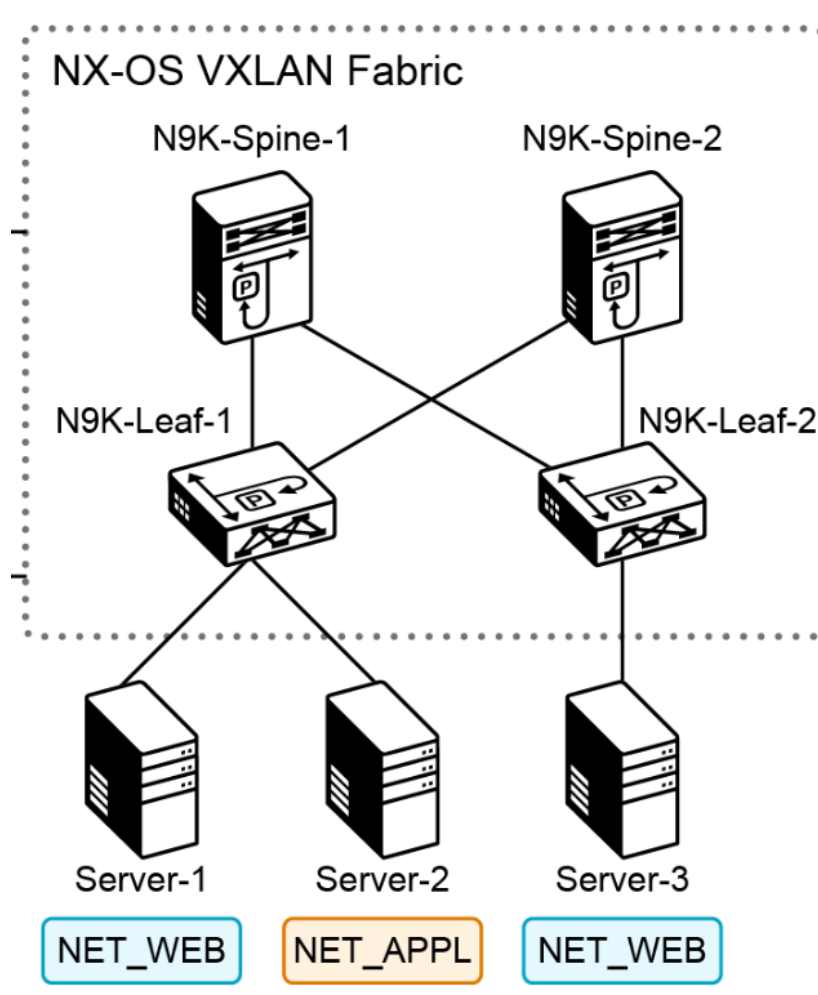
Create a VXLAN Fabric using NDFC

Introduction

This lab demonstrates the fundamentals of creating and configuring a VXLAN EVPN fabric using Cisco NDFC. It begins with fabric creation, followed by switch discovery and onboarding, assignment of spine and leaf roles, and deployment of the initial fabric configuration. Once the fabric nodes are onboarded, the lab demonstrates how to create Virtual Routing and Forwarding (VRF) instances and overlay networks. Finally, it covers interface configuration and the attachment of interfaces to overlay networks to establish end-to-end connectivity between two PCs connected to different leaf switches.

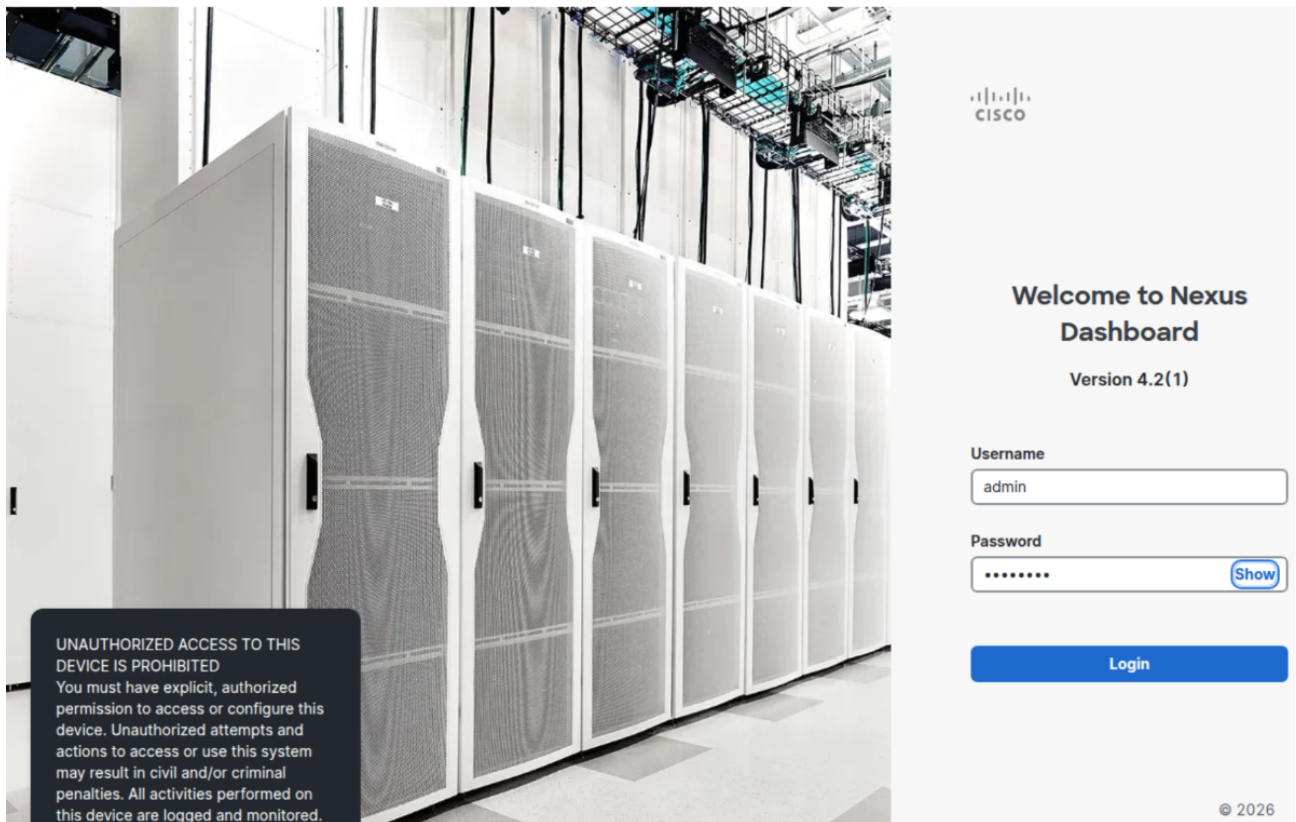
Lab Topology

The lab topology consists of a simple two-spine, two-leaf VXLAN EVPN fabric managed by Cisco NDFC. NDFC provides centralized fabric management and configuration, while each leaf switch connects to both spine switches to form the underlay fabric. **Server-1** and **Server-2** connect to n9k-leaf-01 on the NET_WEB and NET_APP networks respectively, and **Server-3** connects to n9k-leaf-02 on the NET_WEB network. The lab uses this topology to demonstrate fabric onboarding, overlay network creation, interface configuration, and end-to-end connectivity across the VXLAN fabric.

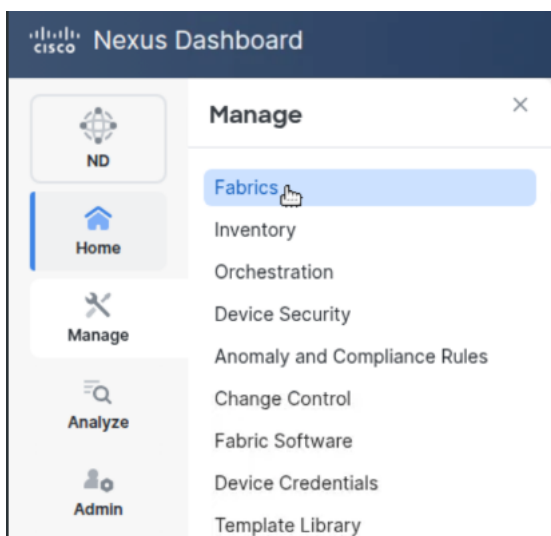


Create a Fabric

Login Nexus Dashboard



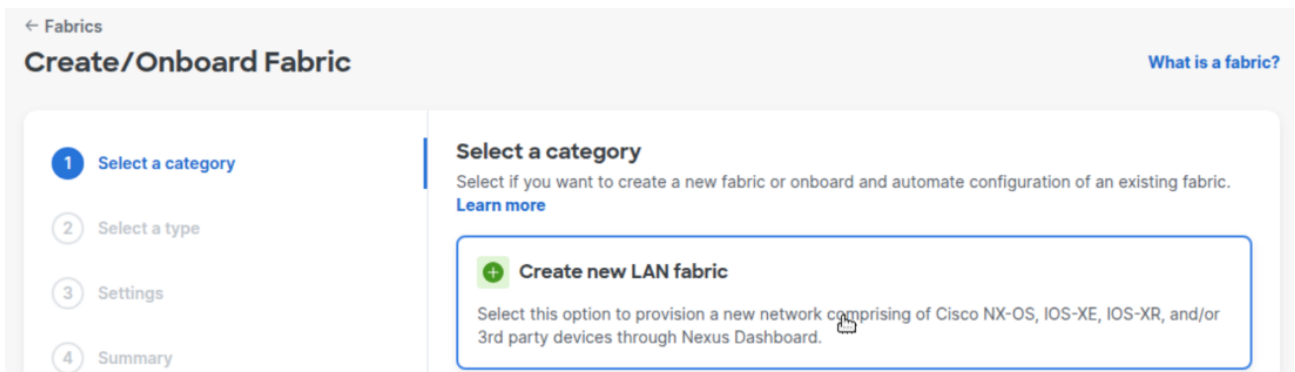
Click **Manage**, then click **Fabrics** in the navigation panel that opens.



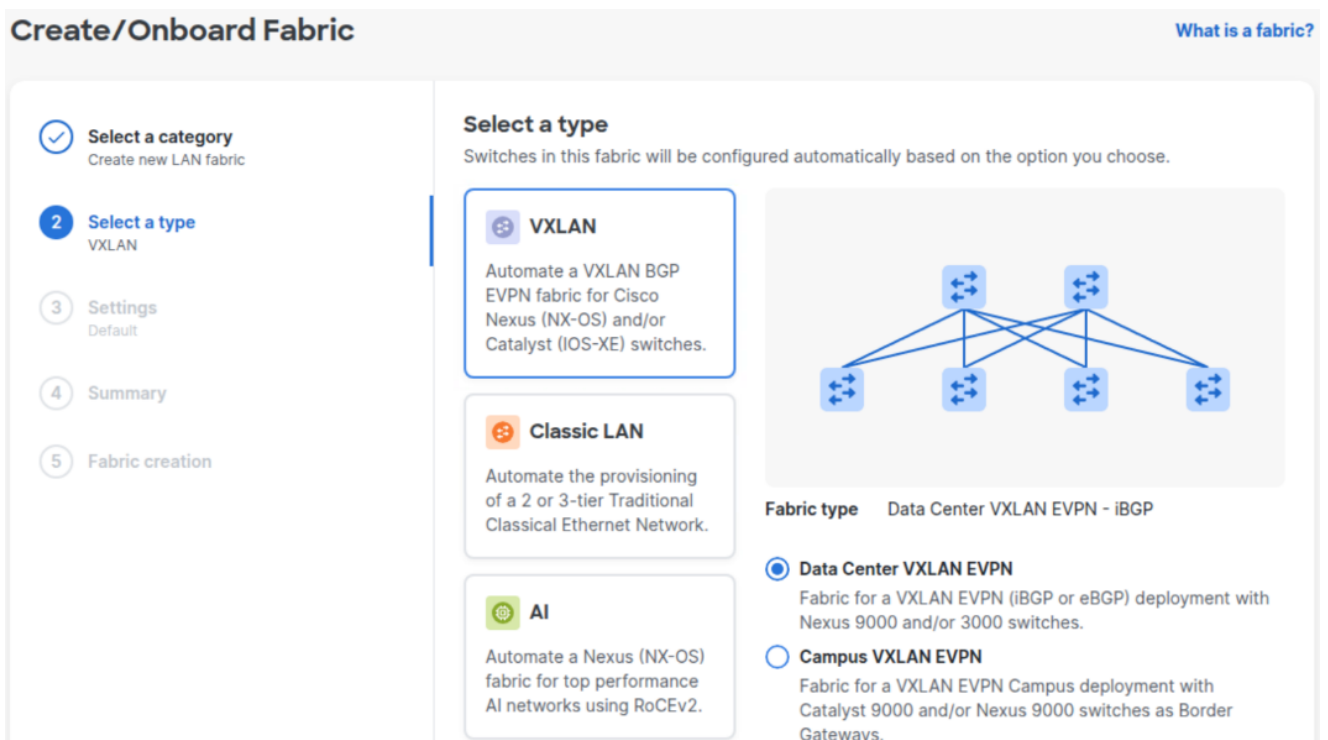
On the **Fabrics** page, click the **Actions** menu in the top right and choose **Create fabric**.



Click the **Create new LAN fabric** tile, then click **Next**.



Click the **VXLAN** tile, leave **Data Center VXLAN EVPN** chosen as the fabric template, then click **Next**.



On the Settings page, change **Configuration mode** from Default to **Advanced**.

Set the fabric **Name** to **OR-TAMBO** and **BGP ASN** to **65001**.

← Fabrics

Create/Onboard Fabric

[What is a fabric?](#)

✓ Select a category
Create new LAN fabric

✓ Select a type
VXLAN

3 Settings
Advanced

4 Advanced settings

5 Summary

6 Fabric creation

Settings

These are the recommended settings for configuring the parameters and capabilities of the new fabric.

Configuration mode ⓘ

☐ Default ☒ Advanced

Name*

OR-TAMBO

Location*

Johannesburg, ZA

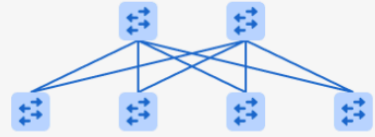
Overlay routing protocol

☒ iBGP ☐ eBGP

BGP ASN*

65001

1-4294967295 | 1-65535[0-65535]



Fabric type Data Center VXLAN EVPN - iBGP

Click **Next**.

Review the **General Parameters**. In this lab, all entries are left as defaults.

✓ Select a category
Create new LAN fabric

✓ Select a type
VXLAN

✓ Settings
Advanced

4 Advanced settings

5 Summary

6 Fabric creation

Advanced settings

The following optional settings will be deployed and/or used when deploying this fabric.

General Parameters Replication vPC Protocols Security Advanced Freeform

☐ **Enable IPv6 Underlay**
If not enabled, IPv4 underlay is used

☐ **Enable IPv6 Link-Local Address**
If not enabled, Spine-Leaf interfaces will use global IPv6 addresses

Fabric Interface Numbering*

p2p

Numbered(Point-to-Point) or Unnumbered

Underlay Subnet IP Mask*

30

Mask for Underlay Subnet IP Range

Underlay Subnet IPv6 Mask

Select an Option

Mask for Underlay Subnet IPv6 Range

Underlay Routing Protocol*

ospf

Used for Spine-Leaf Connectivity

Route-Reflectors*

2

Number of spines acting as Route-Reflectors

Anycast Gateway MAC*

2020.0000.00aa

Click the **Advanced** tab.

Scroll down to **Add Switches without Reload**, open the drop-down, and choose **enable**.

Add Switches without Reload*

enable

Allow switch configuration to be cleared without a reload when Preserve Config is un-checked

By default, Cisco Nexus Dashboard reloads a switch when it imports it with Preserve Config unchecked, to guarantee a clean baseline. Enabling Add Switches without Reload clears the running configuration and applies the calculated configuration without rebooting the switch.

Leave everything else as defaults and click **Next**.

Review the summary and **Submit**.

✓ Select a category
Create new LAN fabric

✓ Select a type
VXLAN

✓ Settings
Advanced

✓ Advanced settings

5 Summary

6 Fabric creation

Summary

Review your selections below.

Category

Fabric category New LAN fabric

Type

Fabric type VXLAN

Fabric sub-type Data Center VXLAN EVPN - iBGP

Settings

Name OR-TAMBO

Location Johannesburg, ZA

Wait for the Fabric creation process to complete.

✓ Select a category
Create new LAN fabric

✓ Select a type
VXLAN

✓ Settings
Advanced

✓ Advanced settings

✓ Summary

6 Fabric creation

Fabric creation

i If you close the session, your fabric will still be created. You can access it any time in the Manage → Fabrics section, where you can add switches to it by clicking on the Actions menu and then on Add Switches.



Creating fabric...



Add Switches and Assign Roles

Under this section, the switches will be onboarded in the VXLAN fabric and will be assigned roles. Click **“Add switches to fabric”**.

Create/Onboard Fabric



All done!

The fabric OR-TAMBO, has successfully been created/onboarded to Nexus Dashboard.

You can modify any of these settings from Manage > Fabrics > Actions by clicking 'Edit fabric settings'

[View fabric details](#)[Add switches to fabric](#)[Create another fabric](#)

You will see a warning that you need to set the credentials for Cisco Nexus Dashboard to access the switches. Click **Set Default Credentials**.



Warning

When changing device configuration, Nexus Dashboard uses the device credentials provided by the user. Do you want to set device credentials now?

[Cancel](#)[Set Default Credentials](#)

Create and save the default credentials.

Set Default Credentials

Username*

admin

Password*

.....

Show

Confirm password*

.....

Show

☐ Robot

Robot selection allows all users with appropriate roles, to push configurations to devices without setting up device credentials.

Cancel

Save

Fill in the Seed IP details and login credentials. **Ensure to untick "Preserve config"**.

Add switches - Fabric: OR-TAMBO

Switch addition mechanism*

☒ Discover

Seed switch details

Seed IP*



Ex: "2.2.2.20" or "10.10.10.40-60" or "2.2.2.20, 2.2.2.21"

Authentication / Privacy*

MD5

Username*

admin

Password*

.....

Show

☐ Set as individual device write credential

Max hops*

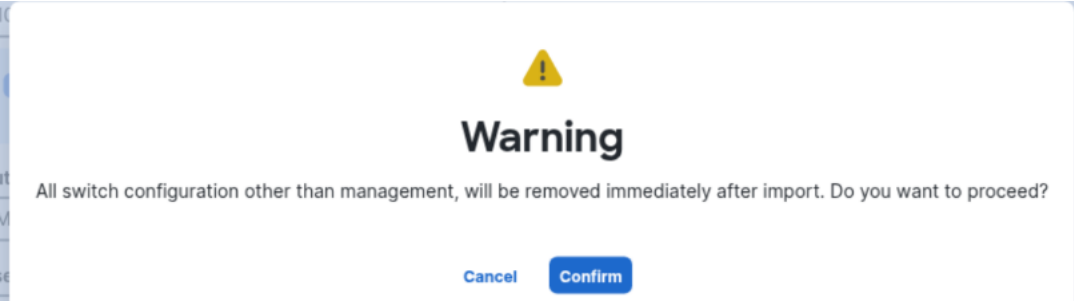
2

☐ Preserve config

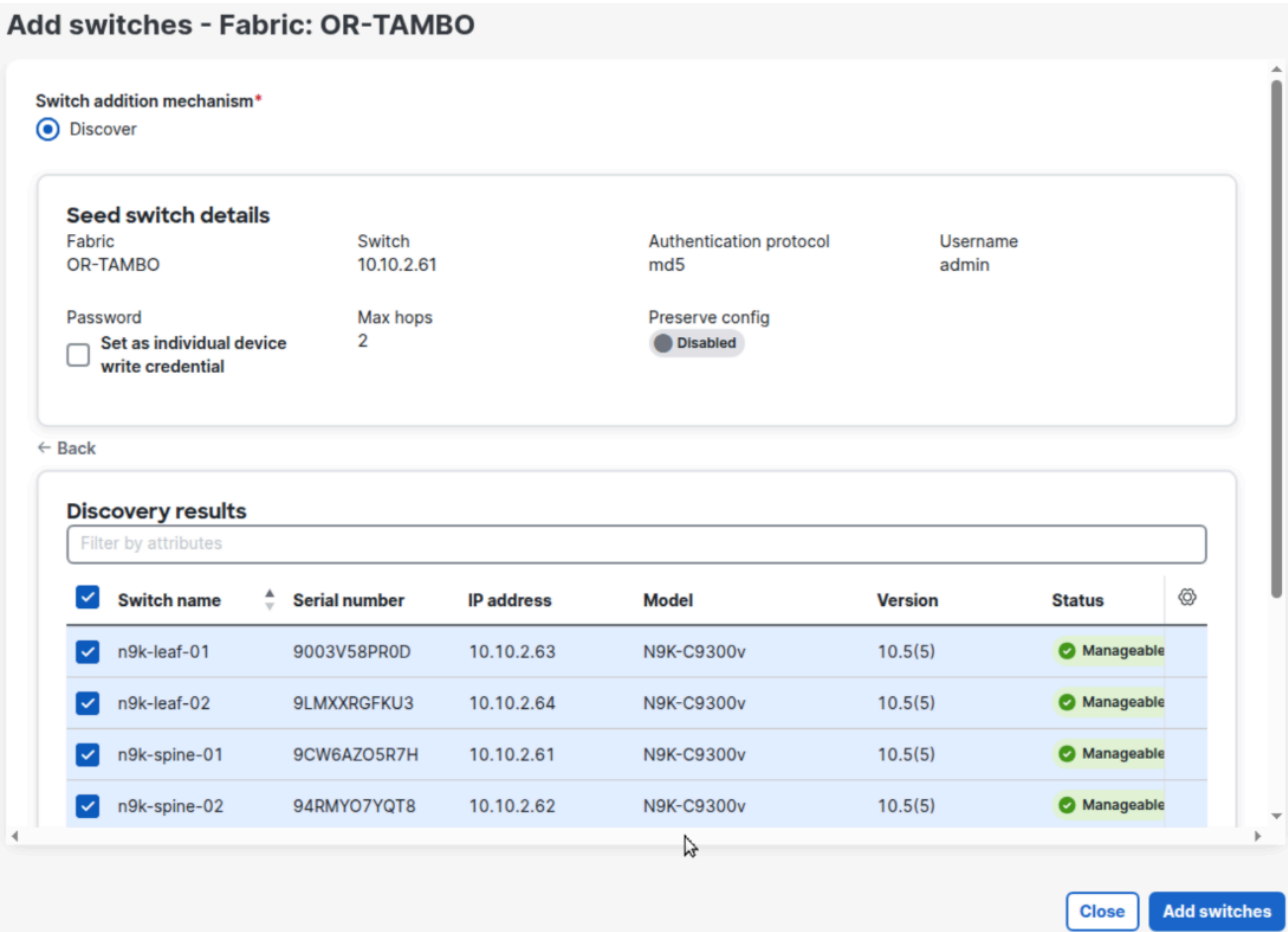


Unchecking this will clean up the configuration on switch(es)

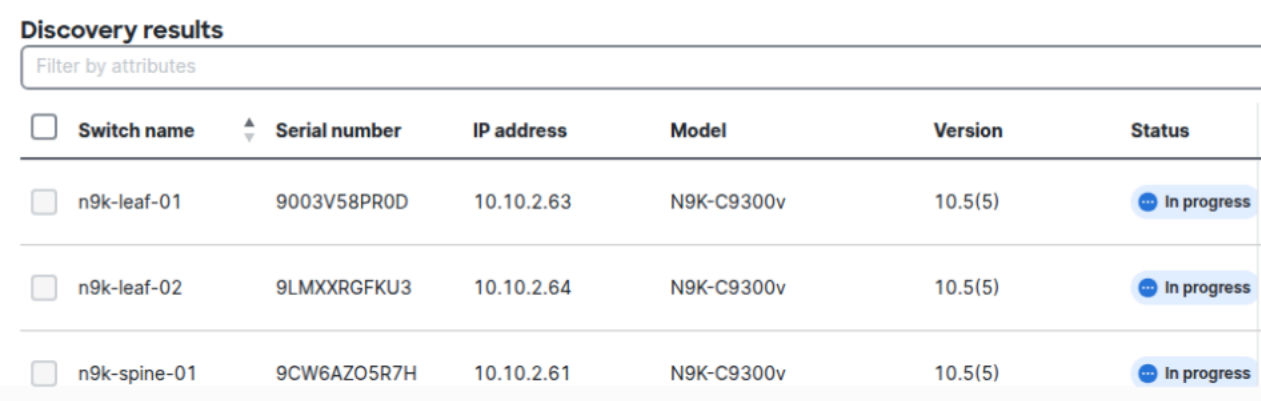
Click Discover switches to begin the switches discovery process. In the cleanup warning dialog, click **Confirm**.



Tick on all the discovered switches and Click on **Add Switches**.



Wait for the discovery process to complete.



Add switches - Fabric: OR-TAMBO

Discovery results

Filter by attributes

☐	Switch name	Serial number	IP address	Model	Version	Status
☐	n9k-leaf-01	9003V58PROD	10.10.2.63	N9K-C9300v	10.5(5)	✓ Switch added
☐	n9k-leaf-02	9LMXXRGFKU3	10.10.2.64	N9K-C9300v	10.5(5)	✓ Switch added
☐	n9k-spine-01	9CW6AZO5R7H	10.10.2.61	N9K-C9300v	10.5(5)	✓ Switch added
☐	n9k-spine-02	94RMYO7YQT8	10.10.2.62	N9K-C9300v	10.5(5)	✓ Switch added

Review the fabric's Inventory page to verify that all devices have been successfully onboarded to the fabric.

OR-TAMBO

Refresh

Overview Inventory Connectivity Segmentation and security Configuration policies Anomalies

Switches VPC pairs Other devices

Filter by attributes

☐	Name	Model	Configuration sync status	Role	Discovery status
☐	n9k-leaf-01	N9K-C9300v	ⓘ Not applicable	Leaf	● Ok
☐	n9k-leaf-02	N9K-C9300v	ⓘ Not applicable	Leaf	● Ok
☐	n9k-spine-01	N9K-C9300v	ⓘ Not applicable	Leaf	● Ok
☐	n9k-spine-02	N9K-C9300v	ⓘ Not applicable	Leaf	● Ok

As observed all devices are by default assigned the role of "Leaf". Check the check boxes for **n9k-spine-01** and **n9k-spine-02** and on the **Actions** menu and choose **Set role**.

Switches VPC pairs Other devices

Filter by attributes

Actions ^

☐	Name	Model	Configuration sync status	Role	Discovery status	
☐	n9k-leaf-01	N9K-C9300v	ⓘ Not applicable	Leaf	● Ok	
☐	n9k-leaf-02	N9K-C9300v	ⓘ Not applicable	Leaf	● Ok	
☒	n9k-spine-01	N9K-C9300v	ⓘ Not applicable	Leaf	● Ok	
☒	n9k-spine-02	N9K-C9300v	ⓘ Not applicable	Leaf	● Ok	

Add switches
Configuration >
Discovery >
Set role
VPC pairing
ToR pairing

Select "Spine" as the role.

Select Role

Search Role

Spine

Leaf (current)

Border

Border Spine

Border Gateway

Border Gateway Spine

Super Spine

In the warning dialog, click **Ok**.



Warning

Please perform "Recalculate and deploy" in the fabric to complete this change prior to "Deploy"

Ok

On the Inventory tab, verify that **n9k-spine-01** and **n9k-spine-02** show role **Spine** and the two leaves remain as **Leaf**.

SwitchesVPC pairsOther devices

Filter by attributes

Name

Model

Configuration sync status

Role

Discovery status

Advisory

n9k-leaf-01

N9K-C9300v

Not applicable

Leaf

Ok

Not ap

n9k-leaf-02

N9K-C9300v

Not applicable

Leaf

Ok

Not ap

n9k-spine-01

N9K-C9300v

Not applicable

Spine

Ok

Not ap

n9k-spine-02

N9K-C9300v

Not applicable

Spine

Ok

Not ap

Click on the top "Action" button and select **Recalculate and deploy**.

← Back

OR-TAMBO

Refresh

View in topology

Actions

Overview

Inventory

Connectivity

Segmentation and security

Configuration policies

Anomalies

Adv

Switches

VPC pairs

Other devices

Edit fabric settings

Add switches

Recalculate and deploy

Associate tenants

Review the **Deploy configuration** preview window, paying attention to the **Pending config** and **Diff** columns for each switch.

Deploy configuration - OR-TAMBO

1

Config preview

2

Deploy progress

Filter by attributes

Resync all

Switch name	Role	Configuration sync status	Pending config	Diff	Status description	Progress	Resync switch	
n9k-spine-02	Spine	Out of sync	382 Lines	+120 -0	Out-of-Sync		Resync	
n9k-spine-01	Spine	Out of sync	382 Lines	+120 -0	Out-of-Sync		Resync	
n9k-leaf-01	Leaf	Out of sync	584 Lines	+382 -62	Out-of-Sync		Resync	
n9k-leaf-02	Leaf	Out of sync	584 Lines	+382 -62	Out-of-Sync		Resync	

Click the **584 Lines** link in the **Pending config** column for **n9k-leaf-01**.

Click the **Side-by-side comparison** tab at the top of the preview to see the configuration that will be added or removed on the switch.

Pending config - OR-TAMBO - n9k-leaf-01

Pending config

Side-by-side comparison

Running config	Expected config
<pre>!Command: show running-config !Running configuration last done at: Tue Sep 1 18:57:39 2026 !Time: Tue Sep 1 19:04:53 2026 boot nxos bootflash:/nxos64-cs-lite.10.5.5.M.bin copp profile strict hostname n9k-leaf-01 interface ethernet1/1</pre>	<pre>!Command: show running-config cfs eth distribute !Running configuration last done at: Tue Sep 1 18:57:39 2026 !Time: Tue Sep 1 19:04:53 2026 boot nxos bootflash:/nxos64-cs-lite.10.5.5.M.bin copp profile strict evpn fabric forwarding anycast-gateway-mac 2020.0000.00aa feature bgp feature dhcp feature interface-vlan feature lacp feature lldp feature ngoam feature nv overlay feature nxapi feature ospf feature pim feature vn-segment-vlan-based hostname n9k-leaf-01 interface ethernet1/1 description connected-to-n9k-spine-01-Ethernet1/1 ip address 10.4.0.1/30 ip ospf network point-to-point</pre>

Click **"Deploy all"** to configure the devices.

Wait for the deployment process to run until completion.

✓ 2
 Config preview Deploy progress

Switch name	IP address	Configuration sync status	Status description	Progress
n9k-spine-02	10.10.2.62	Started	Deployment in progress.	Executed 3 / 382
n9k-spine-01	10.10.2.61	Started	Deployment in progress.	Executed 1 / 382
n9k-leaf-01	10.10.2.63	Started	Deployment in progress.	Executed 2 / 584
n9k-leaf-02	10.10.2.64	Started	Deployment in progress.	Executed 3 / 584

✓ 2
 Config preview Deploy progress

Switch name	IP address	Configuration sync status	Status description	Progress
n9k-spine-02	10.10.2.62	Success	Deployment completed.	Executed 382 / 382
n9k-spine-01	10.10.2.61	Success	Deployment completed.	Executed 382 / 382
n9k-leaf-01	10.10.2.63	Success	Deployment completed.	Executed 584 / 584
n9k-leaf-02	10.10.2.64	Success	Deployment completed.	Executed 584 / 584

Ensure that the **Config-sync status** reads **in sync** for each device.

☐	Name	Model	Configuration sync status	Role	Discovery status
☐	n9k-leaf-01	N9K-C9300v	In sync	Leaf	Ok
☐	n9k-leaf-02	N9K-C9300v	In sync	Leaf	Ok
☐	n9k-spine-01	N9K-C9300v	In sync	Spine	Ok
☐	n9k-spine-02	N9K-C9300v	In sync	Spine	Ok

At this point a Data Center VXLAN EVPN fabric has been successfully created in Cisco Nexus Dashboard, switches successfully onboarded and assigned their respective roles.

Leaf output:

Physical interfaces towards spines are up and loopback interfaces are up.

```
n9k-leaf-02# show ip int br

IP Interface Status for VRF "default"(1)
Interface          IP Address          Interface Status
Lo0                10.2.0.2            protocol-up/link-up/admin-up
Lo1                10.3.0.1            protocol-up/link-up/admin-up
Eth1/1             10.4.0.10           protocol-up/link-up/admin-up
Eth1/2             10.4.0.6            protocol-up/link-up/admin-up
```

Spine output:

Physical interfaces towards leaves are up and loopback interfaces are up.

```
n9k-spine-02# show ip int br

IP Interface Status for VRF "default"(1)
Interface          IP Address          Interface Status
Lo0                10.2.0.1            protocol-up/link-up/admin-up
Lo254              10.254.254.1        protocol-up/link-up/admin-up
Eth1/1             10.4.0.14           protocol-up/link-up/admin-up
Eth1/2             10.4.0.5            protocol-up/link-up/admin-up
```

General **Fabric management** External streaming

General Parameters Replication vPC Protocols Security Advanced Freeform **Resources**

☐ Manual Underlay IP Address Allocation
Checking this will disable Dynamic Underlay IP Address Allocations

Underlay Routing Loopback IP Range*
10.2.0.0/22
Typically Loopback0 IP Address Range

Underlay VTEP Loopback IP Range*
10.3.0.0/22
Typically Loopback1 IP Address Range

Underlay RP Loopback IP Range*
10.254.254.0/24
Anycast or Phantom RP IP Address Range

Underlay Subnet IP Range*
10.4.0.0/16
Address range to assign Numbered and Peer Link SVI IPs

Take note of the IP addresses that were auto-assigned on the devices and the IP Ranges on the Resources page.

Loopback0: underlay routing – **10.2.x.x**

Loopback1: underlay VTEP – **10.3.x.x**

Loopback254: underlay RP – **10.254.254.x**

At this point, the leafs and spines have formed underlay (OSPF) and overlay (BGP L2VPN EVPN) neighborships, even though there are no prefixes yet exchanged.

```
n9k-spine-02# show ip ospf neighbor
OSPF Process ID UNDERLAY VRF default
Total number of neighbors: 2
Neighbor ID      Pri State           Up Time  Address      Interface
10.2.0.3         1 FULL/ -         00:26:11 10.4.0.13    Eth1/1
10.2.0.2         1 FULL/ -         00:26:10 10.4.0.6     Eth1/2
```

```
n9k-spine-02# show bgp l2vpn evpn summary
BGP summary information for VRF default, address family L2VPN EVPN
BGP router identifier 10.2.0.1, local AS number 65001
BGP table version is 3, L2VPN EVPN config peers 2, capable peers 2
0 network entries and 0 paths using 0 bytes of memory
BGP attribute entries [0/0], BGP AS path entries [0/0]
BGP community entries [0/0], BGP clusterlist entries [0/0]

Neighbor      V   AS   MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd
10.2.0.2      4 65001    34       34      3     0    0 00:26:37 0
10.2.0.3      4 65001    34       32      3     0    0 00:26:55 0

Neighbor      T   AS  Type-1   Type-2   Type-3   Type-4   Type-5   Type-6   Type-7   Type-8   Type-12
10.2.0.2      I 65001 0         0         0         0         0         0         0         0         0
10.2.0.3      I 65001 0         0         0         0         0         0         0         0         0
```

Next stop is to create a new VRF and new Networks (Layer 2 and Layer 3).

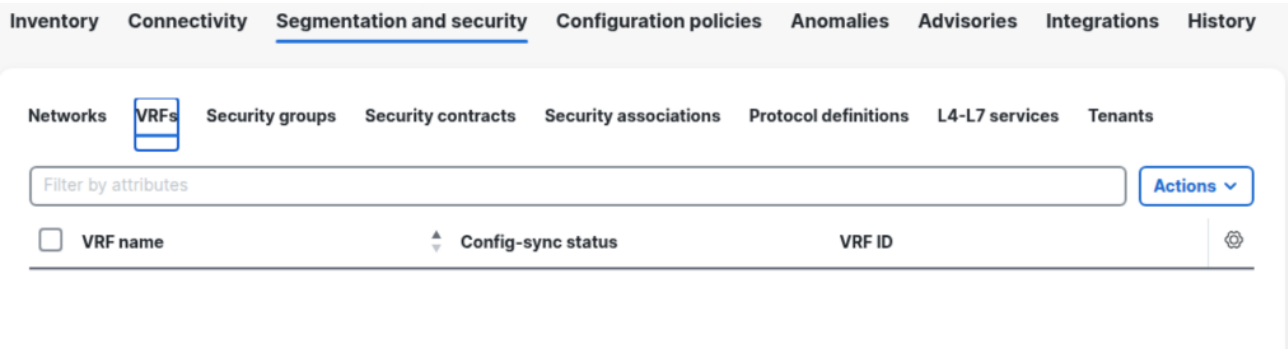
Create a VRF

In Cisco Nexus Dashboard, Virtual Routing and Forwarding (VRF) instances act as isolated, tenant-specific routing tables. The VRF is identified by a Layer 3 VNI (L3VNI) that spans the entire fabric.

Navigate to **Manage > Fabrics**.

From the list of fabrics, click on the fabric to open it.

Click the **Segmentation and security** tab, and then click **VRFs**.



Click **Actions** in the VRF subtab, then **create**.

Filter by attributes

Actions ^

☐ VRF name	▲▼ Config-sync status	VRF ID	Create
			Edit

In the **Create VRF** page, enter **MAIN** for the **VRF name**, keep the proposed VRF ID **50000**, and click **Propose VLAN** to assign the tenant VLAN ID **2000**. Leave all other fields empty or as defaults.

Create VRF

VRF name*

VRF ID*

VLAN ID

 [Propose VLAN](#)

General Parameters Advanced TRM Route Target VRF Lite

VRF VLAN Name

If > 32 chars, enable 'system vlan long-name' for NX-OS, disable VTPv1 and VTPv2 or switch to VTPv3 for IOS XE. Not applicable to L3VNI w/o VLAN config

VRF Interface Description

Not applicable to L3VNI w/o VLAN config

Note: The **VRF ID** is derived from **the Layer 3 VXLAN VNI Range** in the Fabric's Resources and the **VLAN ID** is derived from the **VRF VLAN Range**.

Layer 2 VXLAN VNI Range*

Overlay Network Identifier Range (Min:1, Max:16777214)

Layer 3 VXLAN VNI Range*

Overlay VRF Identifier Range (Min:1, Max:16777214)

Network VLAN Range*

Per Switch Overlay Network VLAN Range (Min:2, Max:4094)

VRF VLAN Range*

Per Switch Overlay VRF VLAN Range (Min:2, Max:4094)

Click **Create** to add the new VRF.

The **MAIN** VRF now appears in the VRFs list.

Inventory

Connectivity

Segmentation and security

Configuration policies

Anomalies

Networks

VRFs

Security groups

Security contracts

Security associations

Protocol definitions

Filter by attributes

☐	VRF name	▲ ▼ Config-sync status	VRF ID
☐	MAIN	Not applicable	50000

Create a Layer 2 Network

In the VXLAN fabric, in the **Segmentation and security** tab, click **Networks**.

Inventory

Connectivity

Segmentation and security

Configuration policies

Anomalies

Advisories

Integrations

History

Networks

VRFs

Security groups

Security contracts

Security associations

Protocol definitions

L4-L7 services

Tenants

Filter by attributes

Actions

☐	Network name	Network ID	VRF name	IPv4 gateway/pr...	IPv6 gateway/pr...	Config-sync status	VLAN ID	VLAN name	Interface group	
--------------------------	--------------	------------	----------	--------------------	--------------------	--------------------	---------	-----------	-----------------	--

Click **Actions** in the **Networks** subtab, then **Create** to create a new network.

Networks

VRFs

Security groups

Security contracts

Security associations

Protocol definitions

L4-L7 services

Tenants

Filter by attributes

Actions ^

☐	Network name	Network ID	VRF name	IPv4 gateway/pr...	IPv6 gateway/pr...	Config-sync status	VLAN ID	VLAN name	Interface group
									Create Edit

Enter **NET_WEB** for the **Network name**, set **Network mode** to **Layer 2 only** (Default is Layer 3), keep the proposed **Network ID** as **30000**, and click **Propose VLAN** to assign VLAN **2300**.

Create network

Network name*

Network mode*

VRF name*

Network ID*

VLAN ID

[Propose VLAN](#)

Leave all other fields as empty or defaults and click **Create**.

Again, just to showcase that the Network ID and VLAN ID are not just random. These values are automatically carved from the reservation pools defined in the Fabric's Resources. The Network ID was derived from the Layer 2 VXLAN VNI Range and the VLAN ID was derived from the Network VLAN Range.

Layer 2 VXLAN VNI Range*

Overlay Network Identifier Range (Min:1, Max:16777214)

Layer 3 VXLAN VNI Range*

Overlay VRF Identifier Range (Min:1, Max:16777214)

Network VLAN Range*

Per Switch Overlay Network VLAN Range (Min:2, Max:4094)

The Layer 2 Network is successfully created as shown below.

Inventory Connectivity <u>Segmentation and security</u> Configuration policies Anomalies Ad						
Networks VRFs Security groups Security contracts Security associations Protocol definitions L						
Filter by attributes						
☐ Network name	Network ID	VRF name	IPv4 gateway/p...	IPv6 gateway/p...	Config-sync status	VLAN ID
☐ NET_WEB	30000	NA			Not applicable	2300

Create a Layer 3 Network

This section showcases how to create a routed Layer 3 network in the MAIN VRF.

On the **Networks** subtab of the **VXLAN** fabric, click **Actions > Create**.

Enter **NET_APP** as the **Network name**, set **Network mode** to **Layer 3**, choose **MAIN** as the **VRF name**, set the **Network ID** to **30001**, and click **Propose VLAN** to assign VLAN **2301**.

In the **IPv4 Gateway/NetMask** field, enter **192.168.2.254/24**.

Create network

Network name*

NET_APP

Network mode*

Layer 3

VRF name*

MAIN

Network ID*

30001

VLAN ID

2301

Propose VLAN

Generate Multicast IP

Please click only to generate a New Multicast Group address and override the default value

General Parameters

Advanced

IPv4 Gateway/NetMask

192.168.2.254/24

example 192.0.2.1/24

Leave the other fields as defaults or empty and click **Create**.

The network **NET_APP** now appears in the **Networks** list, associated with VRF **MAIN** and gateway **192.168.2.254/24**.

Networks								
Filter by attributes								
☐	Network name	Network ID	VRF name	IPv4 gateway/prefix	IPv6 gateway/p...	Config-sync status	VLAN ID	
☐	NET_APP	30001	MAIN	192.168.2.254/24		Not applicable	2301	
☐	NET_WEB	30000	NA			Not applicable	2300	

Click **Actions** in the top-right corner, then **Recalculate and Deploy** to push the configuration to the fabric.

← Back

VXLAN-01

RefreshView in topologyActions

InventoryConnectivitySegmentation and securityConfiguration policiesAnomaliesAdvisories

Edit fabric settings

Add switches

Recalculate and deploy

The Configuration Deployment window shows that there are no configurations to push to the switches. This is expected because the created VRF and networks are not yet attached to the switch interfaces.

Deploy configuration - VXLAN-01

1

2

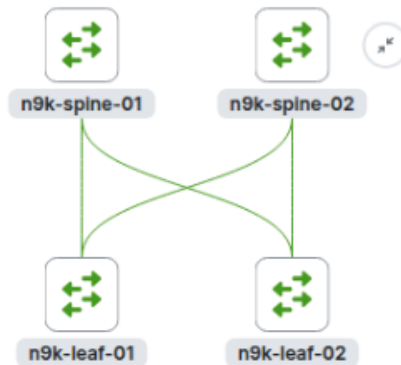
Config previewDeploy progress

Filter by attributes

Resync all

Switch name	Role	Configuration sync status	Pending config	Diff	Status description	Progress	Resync switch	
n9k-leaf-01	Leaf	In sync	0 Lines	+0 -0	In-Sync		Resync	
n9k-spine-02	Spine	In sync	0 Lines	+0 -0	In-Sync		Resync	
n9k-leaf-02	Leaf	In sync	0 Lines	+0 -0	In-Sync		Resync	
n9k-spine-01	Spine	In sync	0 Lines	+0 -0	In-Sync		Resync	

The topology below does indicate that the Fabric now has Networks and VRF objects even though they have not been deployed to the switches.



Attach Endpoints

In VXLAN, attaching an endpoint tells the fabric which network a port belongs to and how to forward its traffic. This section of the lab will show how to set the leaf interface mode, attach the interface to the network and deploy the configurations to the switches.

The host facing ports will be configured to **access mode** in the fabric.

Choose **Manage > Fabrics**.

Click **OR-TAMBO** in the Fabrics list to open the fabric.

On the **Connectivity** tab, open the **Interfaces** view and filter for interface Ethernet1/3 for both leaf switches.

Select the **Ethernet1/3** row on both **n9k-leaf-01** and **n9k-leaf-02** and click the lower **Actions** button, then choose **Edit configuration**.

Inventory <u>Connectivity</u> Segmentation and security Configuration policies Anomalies Advisories Integrations History						
Interfaces Interface groups Links Routing policies Inter-fabric L3 neighbors Endpoints Routes Flows Virtual infrastructure						
Interface == Ethernet1/3 × Edit Clear All Actions ^						
☒	Interface	Switch	Admin status	Operational status	Reason	Intended configuration mode
☒	Ethernet1/3	n9k-leaf-01	↑ Up	↑ Up	ok	trunk
☒	Ethernet1/3	n9k-leaf-02	↑ Up	↑ Up	ok	trunk

Create interface
 Edit configuration
 Configuration >
 Interface group >
 Maintenance >

Open the **Mode** drop-down for **n9k-leaf-01** Ethernet1/3 and select **Access**.

Edit interface(s)

1 of 2 Selected Interface(s) :

Interface

n9k-leaf-01 : Ethernet1/3

Mode*

Access

Policy*

Host

Policy Options

General Parameters

Storm Control

Enable BPDU Guard*

enable

Enable spanning-tree bpduguard: true='enable', false='disable', no='return to default settings'

Configure BPDU Filter

default

Configure spanning-tree bpdufilter, no='return to default settings'

Spanning-tree Link-type

auto

Specify a link type for spanning tree protocol use, default is auto

Scroll down, make sure that the **Access VLAN** box is empty and click **Save & Next**.

Edit interface(s)

☒ Enable Port Type Fast

Enable spanning-tree edge port behavior

MTU*

jumbo

MTU for the Interface

SPEED*

auto

Interface Speed

Access Vlan

VLAN for this access port

Edit interface(s)

2 of 2 Selected Interface(s) :

Interface

n9k-leaf-02 : Ethernet1/3

Mode*

Access

Policy*

Host

Edit interface(s)

☒ Enable Port Type Fast

Enable spanning-tree edge port behavior

MTU*

jumbo

MTU for the Interface

SPEED*

auto

Interface Speed

Access Vlan

VLAN for this access port

Click Save then Click Deploy

Deploy interfaces configuration

1

Config preview

2

Deploy progress

Filter by attributes

Fabric name	Device name	Interface	Admin status	Operation status	Pending config
VXLAN-01	n9k-leaf-01	Ethernet1/3	↑ Up	↑ Up	12 Lines
VXLAN-01	n9k-leaf-02	Ethernet1/3	↑ Up	↑ Up	12 Lines

The pending configuration shows that the interfaces will be configured as access ports, instead of the default (trunk) and spanning-tree based configurations for these host facing ports are added.

Pending config - VXLAN-01 - Ethernet1/3 - n9k-leaf-01

Pending config Side-by-side comparison

```
1 interface ethernet1/3
2   no spanning-tree port type edge trunk
3   no switchport trunk allowed vlan none
4   no switchport mode trunk
5 interface ethernet1/3
6   switchport
7   switchport mode access
8   mtu 9216
9   spanning-tree bpduguard enable
10  spanning-tree port type edge
11  no shutdown
12 configure terminal
```

Click **Deploy Config** to push the change to the switches.

Filter by attributes

Fabric name	Device name	Interface	Admin status	Operation status	Deploy Config
VXLAN-01	n9k-leaf-01	Ethernet1/3	↑ Up	↑ Up	deploying...
VXLAN-01	n9k-leaf-02	Ethernet1/3	↑ Up	↑ Up	deploying...

Confirm that both **Ethernet1/3** interfaces show intended configuration mode **access** and sync status **In-Sync**.

☐	Interface	Switch	Admin status	Operational status	Intended configuration mode	Policy	Sync status
☐	Ethernet1/3	n9k-leaf-01	↑ Up	↑ Up	access	Host	In sync
☐	Ethernet1/3	n9k-leaf-02	↑ Up	↑ Up	access	Host	In sync

Repeat the filter from Step 4, but for **Ethernet1/4** on n9k-leaf-01. Click the lower **Actions** button, then choose **Edit configuration**.

Interfaces

Interface groups

Links

Routing policies

Inter-fabric

L3 neighbors

Endpoints

Routes

Flows

Virtual infrastructure

Interface == Ethernet1/4

Edit

Clear All

Actions

Interface	Switch	Admin status	Operational status	Intended configuration mode	Policy
☒ Ethernet1/4	n9k-leaf-01	↑ Up	↑ Up	trunk	Host

Create interface

Edit configuration

Configuration

Edit interface(s)

Interface

n9k-leaf-01 : Ethernet1/4

Mode*

Access

Policy*

Host

Verify that the **Ethernet1/4** interface has also been set to **Access** mode.

Interfaces

Interface groups

Links

Routing policies

Inter-fabric

L3 neighbors

Endpoints

Routes

Flows

Virtual infrastructure

Interface == Ethernet1/4

Edit

Clear All

Actions

Interface	Switch	Admin status	Operational status	Intended configuration mode	Policy	Sync status
☐ Ethernet1/4	n9k-leaf-01	↑ Up	↑ Up	access	Host	● In sync

Connect Interfaces to Networks

In this section, the Multi-Attach workflow of the Cisco Nexus Dashboard is used to attach endpoints to the networks

Navigate to the **Segmentation and security**.

Select **NET_WEB**, then click on the lower **Actions** button and select **Multi-Attach**.

Inventory

Connectivity

Segmentation and security

Configuration policies

Anomalies

Advisories

Integrations

History

Networks

VRFs

Security groups

Security contracts

Security associations

Protocol definitions

L4-L7 services

Tenants

Filter by attributes

Actions

Network name	Network ID	VRF name	IPv4 gateway/prefix	IPv6 gateway/p...	Config-sync status	VLAN ID
☐ NET_APPL	30001	MAIN	192.168.2.254/24		● Not applica	2301
☒ NET_WEB	30000	NA			● Not applica	2300

Create

Edit

Multi-attach

Multi-detach

Select both leaf switches **n9k-leaf-01** and **n9k-leaf-02**, then click **Next**.

← Back

Multi-Attach of Networks

1

Select Switches

2

Select Interfaces

Select Switches to attach all Selected Networks (1)

Total No. of Attachment : 2

Filter by attributes

☒	Switch	IP Address	Serial Number	Model	Role
☒	n9k-leaf-01	10.10.2.63	9003V58PR0D	N9K-C9300v	Leaf
☒	n9k-leaf-02	10.10.2.64	9LMXXRGFKU3	N9K-C9300v	Leaf

Select the interfaces that you are going to attach to the network **NET_WEB** on the first leaf. Click the **Select Interfaces** button in the row of **n9k-leaf-01**.

Multi-Attach of Networks

✓

Select Switches

2

Select Interfaces

3

Summary

Select Interfaces

Filter by attributes

☐	Network Name	Switch Name	Peer Switch Name	Interfaces List ⓘ	Action
☐	NET_WEB	n9k-leaf-01			Select Interfaces
☐	NET_WEB	n9k-leaf-02			Select Interfaces

Choose **Ethernet1/3** for **n9k-leaf-01**, then click **Save**.

Select Interfaces of n9k-leaf-01 & NET_WEB

Filter by attributes

☒	Interfaces	Switch	Mode
☒	Ethernet1/3	n9k-leaf-01	access

Select the interfaces that you are going to attach to the network **NET_WEB** on the second leaf.
Click the **Select Interfaces** button in the row of **n9k-leaf-02**.

Choose **Ethernet1/3** for **n9k-leaf-02**, then click **Save**.

Multi-Attach of Networks

1

Select Switches

2

Select Interfaces

3

Summary

Select Interfaces

Filter by attributes

☐	Network Name	Switch Name	Peer Switch Name	Interfaces List	Action
☐	NET_WEB	n9k-leaf-01		n9k-leaf-01(eth1/3)	Select Interfaces
☐	NET_WEB	n9k-leaf-02		n9k-leaf-02(eth1/3)	Select Interfaces

Click **Next**

Click Deploy later and Save

1

Select Switches

2

Select Interfaces

3

Summary

Summary

Networks selected
1

Switches selected
2

Network attachments
2

Switch interface association
2

Switch interface de-association
0

☒ Deploy later

☐ Proceed to full switch deploy (recommended)

☐ Proceed to individual network deploy

Select **NET_APPL**, then click **Actions > Multi-Attach**.

Networks

VRFs

Security groups

Security contracts

Security associations

Protocol definitions

L4-L7 services

Tenants

Filter by attributes

Actions

☐	Network name	Network ID	VRF name	IPv4 gateway/prefix	IPv6 gateway/p...	Config-sync status	VLAN ID	
☒	NET_APPL	30001	MAIN	192.168.2.254/24		Not applica	2301	
☐	NET_WEB	30000	NA			Pending	2300	

Create

Edit

Multi-attach

Multi-detach

Select **n9k-leaf-01**, then click **Next**.

Multi-Attach of Networks

1

Select Switches

2

Select Interfaces

Select Switches to attach all Selected Networks (1)

Total No. of Attachment : 1

Filter by attributes

Choose **Ethernet1/4**, then click **Save**.

✓

Select Switches

2

Select Interfaces

3

Summary

Select Interfaces

Filter by attributes

Bulk Paste

Network Name	Switch Name	Peer Switch Name	Interfaces List ⓘ	Action
NET_APPL	n9k-leaf-01		n9k-leaf-01(eth1/4)	Select Interface

Click **Next**

Multi-Attach of Networks

✓

Select Switches

✓

Select Interfaces

3

Summary

Summary

Networks selected

1

Switches selected

1

Network attachments

1

Switch interface association

1

Switch interface de-association

0

Deploy later

Choose **Deploy later**, then click **Save**.

Click the upper **Actions** button, then select **Recalculate and Deploy**.

VXLAN-01

RefreshView in topologyActions

InventoryConnectivitySegmentation and securityConfiguration policiesAnomaliesAdvisories

NetworksVRFsSecurity groupsSecurity contractsSecurity associationsProtocol definitionsL4-L7 services

Filter by attributes

☐	Network name	Network ID	VRF name	IPv4 gateway/prefix	IPv6 gateway/p...	Config-sync status	VLAN ID	VLAN
☐	NET_APPL	30001	MAIN	192.168.2.254/24		Pending	2301	
☐	NET_WEB	30000	NA			Pending	2300	

Edit fabric settings

Add switches

Recalculate and deploy

Associate tenants

Configuration

Monitoring

Maintenance

Review the deployment intent

12

Config previewDeploy progress

Filter by attributesResync all

Switch name	Role	Configuration sync status	Pending config	Diff	Status description	Progress	Resync switch
n9k-spine-02	Spine	In sync	0 Lines	+0 -0	In-Sync		Resync
n9k-spine-01	Spine	In sync	0 Lines	+0 -0	In-Sync		Resync
n9k-leaf-01	Leaf	Out of sync	81 Lines	+52 -0	Out-of-Sync		Resync
n9k-leaf-02	Leaf	Out of sync	22 Lines	+9 -0	Out-of-Sync		Resync

Pending configSide-by-side comparison

```
vlan 2000
  vn-segment 50000
  configure terminal
  vrf context main
  vni 50000
  rd auto
  address-family ipv4 unicast
    route-target both auto
  route-target both auto evpn
  address-family ipv6 unicast
    route-target both auto
  route-target both auto evpn
exit
interface Vlan2000
  vrf member main
  ip forward
  ipv6 address use-link-local-only
  no ip redirects
  no ipv6 redirects
  ntu 9216
  no shutdown
configure terminal
router bgp 65001
  vrf main
    address-family ipv4 unicast
      advertise l2vpn evpn
      redistribute direct route-map fabric-rmap-redist-subnet
      maximum-paths ibgp 2
    exit
    address-family ipv6 unicast
      advertise l2vpn evpn
      redistribute direct route-map fabric-rmap-redist-subnet
      maximum-paths ibgp 2
  configure terminal
```

```
configure terminal
interface nve1
  member vni 50000 associate-vrf
  member vni 30000
  mcast-group 239.1.1.0
  member vni 30001
  mcast-group 239.1.1.0
configure terminal
vlan 2300
  vn-segment 30000
configure terminal
vlan 2301
  vn-segment 30001
configure terminal
interface Vlan2301
  vrf member main
  ip address 192.168.2.254/24 tag 12345
  fabric forwarding mode anycast-gateway
  no shutdown
configure terminal
configure terminal
evpn
  vni 30000 l2
    rd auto
    route-target import auto
    route-target export auto
  vni 30001 l2
    rd auto
    route-target import auto
    route-target export auto
```

Pending config Side-by-side comparison

```
vn-segment 30001
configure terminal
interface Vlan2301
vrf member main
ip address 192.168.2.254/24 tag 12345
fabric forwarding mode anycast-gateway
no shutdown
configure terminal
configure terminal
evpn
vni 30000 l2
rd auto
route-target import auto
route-target export auto
vni 30001 l2
rd auto
route-target import auto
route-target export auto
interface ethernet1/3
switchport
switchport mode access
mtu 9216
spanning-tree bpduguard enable
spanning-tree port type edge
no shutdown
switchport access vlan 2300
configure terminal
interface ethernet1/4
switchport
switchport mode access
mtu 9216
spanning-tree bpduguard enable
spanning-tree port type edge
no shutdown
switchport access vlan 2301
configure terminal
```

Pending config - VXLAN-01 - n9k-leaf-02

Pending config Side-by-side comparison

```
vlan 2300
vn-segment 30000
configure terminal
interface nve1
member vni 30000
mcast-group 239.1.1.0
configure terminal
configure terminal
evpn
vni 30000 l2
rd auto
route-target import auto
route-target export auto
interface ethernet1/3
switchport
switchport mode access
mtu 9216
spanning-tree bpduguard enable
spanning-tree port type edge
no shutdown
switchport access vlan 2300
configure terminal
```

Confirm that both **NET_WEB** and **NET_APPL** show the **Deployed** status.

Inventory Connectivity <u>Segmentation and security</u> Configuration policies Anomalies Advisories I								
Networks VRFs Security groups Security contracts Security associations Protocol definitions L4-L7 services								
Filter by attributes								
☐	Network name	Network ID	VRF name	IPv4 gateway/prefix	IPv6 gateway/p...	Config-sync status	VLAN ID	VLAN
☐	NET_APPL	30001	MAIN	192.168.2.254/24		Deployed	2301	
☐	NET_WEB	30000	NA			Deployed	2300	

Connect to **Server-1**

From **Server-1 (IP: 192.168.1.101)**, ping **Server-3** at **192.168.1.103**.

```
3: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:0b:22:8d brd ff:ff:ff:ff:ff:ff
    altname enp11s0
    inet 192.168.1.101/24 brd 192.168.1.255 scope global ens192
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe0b:228d/64 scope link
        valid_lft forever preferred_lft forever
student@server1:~$
```

```
student@server1:~$ ping 192.168.1.103
PING 192.168.1.103 (192.168.1.103) 56(84) bytes of data:
64 bytes from 192.168.1.103: icmp_seq=3 ttl=64 time=8.81 ms
64 bytes from 192.168.1.103: icmp_seq=1 ttl=64 time=2048 ms
64 bytes from 192.168.1.103: icmp_seq=2 ttl=64 time=1033 ms
64 bytes from 192.168.1.103: icmp_seq=4 ttl=64 time=3.87 ms
64 bytes from 192.168.1.103: icmp_seq=5 ttl=64 time=3.87 ms
64 bytes from 192.168.1.103: icmp_seq=6 ttl=64 time=3.71 ms
64 bytes from 192.168.1.103: icmp_seq=7 ttl=64 time=3.65 ms
^C
--- 192.168.1.103 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6047ms
rtt min/avg/max/mdev = 3.650/443.537/2047.992/744.900 ms, pipe 3
student@server1:~$
```

From **Server-1**, ping **Server-2** at **192.168.2.102**,

```
student@server1:~$ ping 192.168.2.102
PING 192.168.2.102 (192.168.2.102) 56(84) bytes of data:
From 192.168.1.101 icmp_seq=1 Destination Host Unreachable
From 192.168.1.101 icmp_seq=2 Destination Host Unreachable
From 192.168.1.101 icmp_seq=3 Destination Host Unreachable
From 192.168.1.101 icmp_seq=4 Destination Host Unreachable
From 192.168.1.101 icmp_seq=5 Destination Host Unreachable
From 192.168.1.101 icmp_seq=6 Destination Host Unreachable
^C
--- 192.168.2.102 ping statistics ---
8 packets transmitted, 0 received, +6 errors, 100% packet loss, time 7170ms
pipe 4
student@server1:~$
```

The ping fails with no replies and 100% packet loss. NET_WEB is a Layer 2-only network, so the endpoint reaches other endpoints in the same network but cannot reach another network without a gateway.

From **Server-2**, ping the gateway **192.168.2.254**.

```
student@server2:~$ ping 192.168.2.254
PING 192.168.2.254 (192.168.2.254) 56(84) bytes of data:
64 bytes from 192.168.2.254: icmp_seq=2 ttl=255 time=0.867 ms
64 bytes from 192.168.2.254: icmp_seq=3 ttl=255 time=0.853 ms
64 bytes from 192.168.2.254: icmp_seq=4 ttl=255 time=0.948 ms
^C
--- 192.168.2.254 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3066ms
rtt min/avg/max/mdev = 0.853/0.889/0.948/0.041 ms
student@server2:~$
```

Server 2 can communicate with its default gateway.

Change Network Type from Layer 2 to Layer 3

NET_WEB is a Layer 2 network, so its endpoints reach each other in the same subnet but cannot reach other subnets. In this section of the lab the NET_WEB network will be converted to Layer 3 by adding a distributed anycast gateway (DAG). Inter-subnet connectivity will then be verified.

Navigate to **Segmentation and security > Networks**, select **NET_WEB**, and click **Actions > Edit**.

Change the network mode to **Layer 3** and associate the network with VRF – MAIN

The image shows two side-by-side 'Edit network' forms. The left form shows the initial state: Network name is 'NET_WEB', Network mode is 'Layer 2 only', VRF name is 'NA', Network ID is '30000', and VLAN ID is '2300'. The right form shows the updated state: Network name is 'NET_WEB', Network mode is 'Layer 3', VRF name is 'MAIN', Network ID is '30000', and VLAN ID is '2300'. Blue arrows indicate the changes from 'Layer 2 only' to 'Layer 3' and from 'NA' to 'MAIN'. Both forms have a 'Propose VLAN' button next to the VLAN ID field.

Enter the IPv4 gateway **192.168.1.254/24** for NET_WEB, then click **Save** to convert the Layer 2 network to Layer 3.

The image shows a detailed 'Edit network' form for 'NET_WEB'. The form includes fields for Network name, Network mode (set to 'Layer 3'), VRF name (set to 'MAIN'), Network ID, and VLAN ID. Below these fields is a 'Generate Multicast IP' button and a note: 'Please click only to generate a New Multicast Group address and override the default value'. At the bottom, there are tabs for 'General Parameters' and 'Advanced'. Under 'General Parameters', the 'IPv4 Gateway/NetMask' field is highlighted with a blue border and contains the value '192.168.1.254/24'. Below this field is a small text example: 'example 192.0.2.1/24'.

Select **NET_WEB**, then click **Actions > Deploy**.

Filter by attributes

Actions ^

☐ Network name	Network ID	VRF name	IPv4 gateway/prefix	IPv6 gateway/p...	Config-sync status	VLAN ID	
☐ NET_APPL	30001	MAIN	192.168.2.254/24		Deployed	2301	Create
☒ NET_WEB	30000	MAIN	192.168.1.254/24		Pending	2300	Edit
							Multi-attach
							Multi-detach
							Deploy

Connect to **Server-1** and ping the default gateway.

```
student@server1:~$ ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data:
64 bytes from 192.168.1.254: icmp_seq=2 ttl=255 time=0.843 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=255 time=0.934 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=255 time=0.736 ms
^C
--- 192.168.1.254 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3023ms
rtt min/avg/max/mdev = 0.736/0.837/0.934/0.080 ms
student@server1:~$
```

From **Server-1**, ping **Server-2** at 192.168.2.102.

```
student@server1:~$ ping 192.168.2.102
PING 192.168.2.102 (192.168.2.102) 56(84) bytes of data:
64 bytes from 192.168.2.102: icmp_seq=1 ttl=63 time=1.30 ms
64 bytes from 192.168.2.102: icmp_seq=2 ttl=63 time=1.25 ms
64 bytes from 192.168.2.102: icmp_seq=3 ttl=63 time=1.22 ms
64 bytes from 192.168.2.102: icmp_seq=4 ttl=63 time=1.36 ms
64 bytes from 192.168.2.102: icmp_seq=5 ttl=63 time=1.34 ms
```

Verify EVPN Routes

You verify the Virtual Extensible LAN (VXLAN) Ethernet VPN (EVPN) control plane on a Cisco Nexus 9000 leaf. You read the Border Gateway Protocol (BGP) Layer 2 VPN (L2VPN) EVPN route table and trace one route into the data plane.

On n9k-leaf-02 Run the `show bgp l2vpn evpn summary` command to list the EVPN peers and their route-type counts.

```
n9k-leaf-02# show bgp l2vpn evpn summary
BGP summary information for VRF default, address family L2VPN EVPN
BGP router identifier 10.2.0.2, local AS number 65001
BGP table version is 20, L2VPN EVPN config peers 2, capable peers 2
12 network entries and 18 paths using 4368 bytes of memory
BGP attribute entries [15/5520], BGP AS path entries [0/0]
BGP community entries [0/0], BGP clusterlist entries [2/8]

Neighbor      V    AS    MsgRcvd    MsgSent    TblVer    InQ  OutQ  Up/Down    State/PfxRcd
10.2.0.1      4  65001      81         70         20     0    0 01:01:25 5
10.2.0.4      4  65001      81         68         20     0    0 01:01:56 5

Neighbor      T    AS  Type-1    Type-2    Type-3    Type-4    Type-5    Type-6    Type-7    Type-8    Type-12
10.2.0.1      I  65001  0         3         0         0         2         0         0         0         0
10.2.0.4      I  65001  0         3         0         0         2         0         0         0         0
```

The first table lists the two internal BGP (iBGP) sessions to the spine route reflectors, both in the same autonomous system as this leaf. The second table breaks the received prefixes down by EVPN route type. Type-2 carries host media access control (MAC) and MAC+IP addresses. Type-3 carries broadcast, unknown-unicast, and multicast (BUM) flooding. Type-4 carries Ethernet segment routes for multihoming. Type-5 carries IP prefixes.

Run the `show bgp l2vpn evpn` command to display the full EVPN route table.

```
n9k-leaf-02# show bgp l2vpn evpn
BGP routing table information for VRF default, address family L2VPN EVPN
BGP table version is 20, Local Router ID is 10.2.0.2
Status: s-suppressed, x-deleted, S-stale, d-dampened, h-history, *-valid, >-best
Path type: i-internal, e-external, c-confed, l-local, a-aggregate, r-redist, I-injected
Origin codes: i - IGP, e - EGP, ? - incomplete, | - multipath, & - backup, 2 - best2
```

Network	Next Hop	Metric	LocPrf	Weight	Path
Route Distinguisher: 10.2.0.2:35067 (L2VNI 30000)					
*>l[2]:[0]:[0]:[48]:[000c.2900.18a9]:[0]:[0.0.0.0]/216	10.3.0.1		100	32768	i
*>i[2]:[0]:[0]:[48]:[000c.290b.228d]:[0]:[0.0.0.0]/216	10.3.0.2		100	0	i
*>i[2]:[0]:[0]:[48]:[000c.290b.228d]:[32]:[192.168.1.101]/272	10.3.0.2		100	0	i
Route Distinguisher: 10.2.0.3:4					
* i[5]:[0]:[0]:[24]:[192.168.1.0]/224	10.3.0.2	0	100	0	?
*>i	10.3.0.2	0	100	0	?
* i[5]:[0]:[0]:[24]:[192.168.2.0]/224	10.3.0.2	0	100	0	?
*>i	10.3.0.2	0	100	0	?
Route Distinguisher: 10.2.0.3:35067					
*>i[2]:[0]:[0]:[48]:[000c.290b.228d]:[0]:[0.0.0.0]/216	10.3.0.2		100	0	i
* i	10.3.0.2		100	0	i
*>i[2]:[0]:[0]:[48]:[000c.290b.228d]:[32]:[192.168.1.101]/272	10.3.0.2		100	0	i
* i	10.3.0.2		100	0	i
Route Distinguisher: 10.2.0.3:35068					
* i[2]:[0]:[0]:[48]:[3ea2.cdb8.1048]:[32]:[192.168.2.102]/272	10.3.0.2		100	0	i
*>i	10.3.0.2		100	0	i
Route Distinguisher: 10.2.0.2:4 (L3VNI 50000)					
*>i[2]:[0]:[0]:[48]:[000c.290b.228d]:[32]:[192.168.1.101]/272	10.3.0.2		100	0	i
*>i[2]:[0]:[0]:[48]:[3ea2.cdb8.1048]:[32]:[192.168.2.102]/272	10.3.0.2		100	0	i
* i[5]:[0]:[0]:[24]:[192.168.1.0]/224	10.3.0.2	0	100	0	?
*>l	10.3.0.1	0	100	32768	?
*>i[5]:[0]:[0]:[24]:[192.168.2.0]/224	10.3.0.2	0	100	0	?

The table groups routes into one block per route distinguisher. The RD encodes the router ID of the originating leaf plus a locally significant index, while the inline (**L2VNI 30000**) or (**L3VNI 50000**) tag gives the VXLAN network identifier (VNI). Routes that this leaf originates show path type **l** (local), routes learned from the route reflectors show path type **i** (internal), and the Next Hop column is always the VTEP that owns the route.

The route `[2]:[0]:[0]:[48]:[000c.290b.228d]:[32]:[192.168.1.101]` carries the MAC and IP of Server-1. Its path type is i (internal), the local preference is 100, and the next hop is 10.3.0.2, the VTEP on the remote leaf n9k-leaf-01. This entry proves that n9k-leaf-02 learned the remote endpoint through the EVPN control plane rather than through data-plane flooding.

The route `[2]:[0]:[0]:[48]:[3ea2.cdb8.1048]:[32]:[192.168.2.102]` carries the MAC and IP of Server-2. Its path type is i (internal), the local preference is 100, and the next hop is 10.3.0.2, the VTEP on the remote leaf that Server-2 connects to. n9k-leaf-02 learned this endpoint in the NET_APPL network through the EVPN control plane.

The entries under L3VNI 50000 are Type-5 routes. They advertise IP prefixes rather than individual hosts and provide reachability for the subnets carried in the MAIN VRF instance, identified by the Layer 3 VNI 50000. The next hop 10.3.0.2 is the VTEP through which n9k-leaf-02 reaches those subnets with symmetric IRB.

Run the `show ip route vrf MAIN` command to see how the EVPN routes map to the data plane.

```
n9k-leaf-02# show ip route vrf MAIN
IP Route Table for VRF "main"
'*' denotes best ucast next-hop
*** denotes best mcast next-hop
'[x/y]' denotes [preference/metric]
'%<string>' in via output denotes VRF <string>

192.168.1.0/24, ubest/mbest: 1/0, attached
  *via 192.168.1.254, Vlan2300, [0/0], 00:07:09, direct, tag 12345
192.168.1.101/32, ubest/mbest: 1/0
  *via 10.3.0.2%default, [200/0], 00:06:10, bgp-65001, internal, tag 65001, segid: 50000 tunnelid: 0xa030002 encap: VXLAN
192.168.1.254/32, ubest/mbest: 1/0, attached
  *via 192.168.1.254, Vlan2300, [0/0], 00:07:09, local, tag 12345
192.168.2.0/24, ubest/mbest: 1/0
  *via 10.3.0.2%default, [200/0], 00:06:44, bgp-65001, internal, tag 65001, segid: 50000 tunnelid: 0xa030002 encap: VXLAN
192.168.2.102/32, ubest/mbest: 1/0
  *via 10.3.0.2%default, [200/0], 00:06:44, bgp-65001, internal, tag 65001, segid: 50000 tunnelid: 0xa030002 encap: VXLAN
```

The local subnet 192.168.1.0/24 and its DAG address 192.168.1.254 are **attached** through the Vlan2300 SVI. The remote endpoints and the NET_APPL subnet resolve through the VTEP 10.3.0.2 as **bgp internal** routes. Each carry **segid: 50000**, the Layer 3 VNI of the MAIN VRF, and is encapsulated in VXLAN.

The prefixes have different lengths: a /24 prefix represents a learned network, and a /32 prefix represents a learned host (server). Notice the 192.168.2.0/24 network: it is not learned locally as attached, because this leaf does not have any interfaces in that network.

References

Cisco U Courses:

1. Cisco Data Center Nexus Dashboard Essentials | DCNDE